[BUG] Jacobian with diff_method='best' and 'adjoint' mismatches for defau

https://github.com/PennyLaneAI/pennylane/issues/2273

ROW 117

[BUG] Jacobian with diff_method='best' and 'adjoint' mismatches for default.qubit #2273

Closed #2274

New issue

chaeyeunpark opened on Mar 3, 2022 · edited by chaeyeunpark

Edits

Expected behavior

The following code should work

Actual behavior

Raises an error

Additional information

No response

Source code

```
import pennylane as qml
import pennylane.numpy as np
import pytest
from scipy.stats import unitary_group

def circuit_ansatz(params, wires):
    """Circuit ansatz containing all the parametrized gates"""
    qml.QubitStateVector(unitary_group.rvs(2**4, random_state=0)[0], wires=wires)
    qml.CRX(params[0], wires=[wires[3], wires[0]])

def test():
    dev_def1 = qml.device("default.qubit", wires=range(4))
    dev_def2 = qml.device("default.qubit", wires=range(4))

    def circuit(params):
        circuit_ansatz(params, wires=range(4))
        return qml.expval(qml.PauliX(0)), qml.expval(qml.PauliY(1))

    params = np.array([1.0])

    qnode_def = qml.QNode(circuit, dev_def1)
    qnode_adj = qml.QNode(circuit, dev_def2, diff_method="adjoint")

    j_def = qml.jacobian(qnode_def)(params)
    j_adj = qml.jacobian(qnode_adj)(params)

    print(j_def - j_adj)

    assert np.allclose(j_def, j_adj)

if __name__ == '__main__':
    test()
```

Tracebacks

No response

System information

```
Name: PennyLane
Version: 0.22.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /home/chae-yeun/work/Xanadu/pennylane
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar
Required-by: PennyLane-Lightning, PennyLane-Qchem
Platform info:           Linux-5.13.0-1029-oem-x86_64-with-glibc2.29
Python version:          3.8.10
Numpy version:           1.21.0
Scipy version:           1.7.0
```

Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Fixes the operation derivative
PennyLaneAI/pennylane

Notifications Customize

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Participants

Visu

```
Platform info:           Linux-5.13.0-1029-oem-x86_64-with-glibc2.29
Python version:          3.8.10
Numpy version:           1.21.0
Scipy version:           1.7.0
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.22.0.dev13)
- default.gaussian (PennyLane-0.22.0.dev0)
- default.mixed (PennyLane-0.22.0.dev0)
- default.qubit (PennyLane-0.22.0.dev0)
- default.qubit.autograd (PennyLane-0.22.0.dev0)
- default.qubit.jax (PennyLane-0.22.0.dev0)
- default.qubit.tf (PennyLane-0.22.0.dev0)
- default.qubit.torch (PennyLane-0.22.0.dev0)
```

Existing GitHub issues

- [x] I have searched existing GitHub issues to make sure the issue does not already exist.

chaeyeunpark added bug on Mar 3, 2022

chaeyeunpark changed the title ~~[BUG] Jacobian with diff_method='parameter-shift' and 'adjoint' mismatches for default.qubit~~ [BUG] Jacobian with diff_method='best' and 'adjoint' mismatches for default.qubit on Mar 3, 2022

josh146 on Mar 3, 2022 · edited by josh146 — Edits · Member

Interesting 🤔 It looks like the adjoint might be incorrect here, if you also compare parameter-shift, it agrees with backprop.

Edit: yep, the adjoint method for `default.qubit` is returning the incorrect result.

josh146 on Mar 3, 2022 — Member

Also it doesn't appear in v0.21, so this is a bug in master!

chaeyeunpark on Mar 3, 2022 · edited by chaeyeunpark — Edits · Contributor · Author

@josh146 Thanks Josh. This code is in fact from `Lightning` 's tests, which worked well yesterday. A recent commit #2256 must have entered this bug.

josh146 on Mar 3, 2022 · edited by josh146 — Edits · Member

@chaeyeunpark the issue seems to be with the `qml.matrix()` function. E.g., consider

```
>>> op = qml.CRX(0.5, wires=[3, 0])
>>> H = qml.generator(op, format="observable")
>>> H.ops[0]
Projector([1], wires=[3]) @ PauliX(wires=[0])
>>> H.ops[0].get_matrix()
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 1.]
 [0. 0. 1. 0.]]
>>> qml.matrix(H.ops[0])
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 1.]
 [0. 0. 1. 0.]]
>>> qml.matrix(H)
[[ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]
```

chaeyeunpark on Mar 3, 2022 · edited by chaeyeunpark — Edits · Contributor · Author

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 [0. 0. 0. 1.]
 [0. 0. 1. 0.]]
>>> qml.matrix(H.ops[0])
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 1.]
 [0. 0. 1. 0.]]
>>> qml.matrix(H)
[[ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]
 [ 0. +0.j  0. +0.j  0. +0.j -0.5+0.j]
 [ 0. +0.j  0. +0.j  0. +0.j  0. +0.j]
 [ 0. +0.j -0.5+0.j  0. +0.j  0. +0.j]]
```

For some reason, `qml.matrix(H)` is sorting by wires when it shouldn't be

josh146 linked a pull request that will close this issue Fixes the operation derivative #2274 on Mar 3, 2022

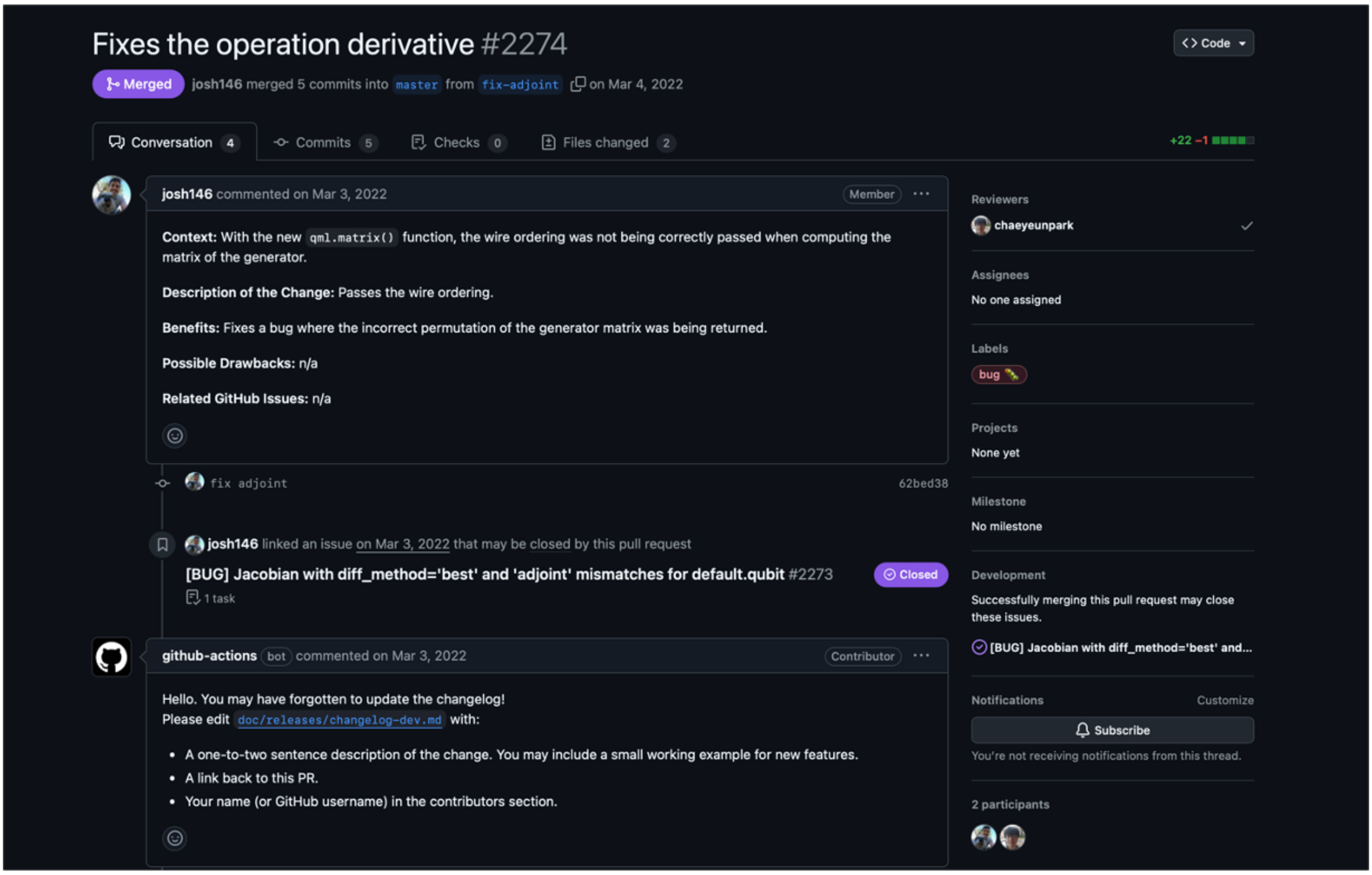
josh146 on Mar 3, 2022 — Member

Hey @chaeyeunpark, could you check if #2273 fixes the issue on the lightning tests? The branch name is `fix-adjoint`.

chaeyeunpark on Mar 3, 2022 — Contributor · Author

Hi @josh146, thanks for your agile response. Yes, that fixes the issue.

josh146 closed this as completed in #2274 on Mar 4, 2022


Fixes the operation derivative #2274
Code
Merged
josh146 merged 5 commits into master from fix-adjoint on Mar 4, 2022
Conversation 4
Commits 5
Checks 0
Files changed 2
+22 −1
josh146 commented on Mar 3, 2022
Member
Context: With the new qml.matrix() function, the wire ordering was not being correctly passed when computing the matrix of the generator.
Description of the Change: Passes the wire ordering.
Benefits: Fixes a bug where the incorrect permutation of the generator matrix was being returned.
Possible Drawbacks: n/a
Related GitHub Issues: n/a
fix adjoint
62bed38
josh146 linked an issue on Mar 3, 2022 that may be closed by this pull request
[BUG] Jacobian with diff_method='best' and 'adjoint' mismatches for default.qubit #2273
Closed
1 task
github-actions bot commented on Mar 3, 2022
Contributor
Hello. You may have forgotten to update the changelog!
Please edit doc/releases/changelog-dev.md with:
A one-to-two sentence description of the change. You may include a small working example for new features.
A link back to this PR.
Your name (or GitHub username) in the contributors section.
Reviewers
chaeyeunpark
Assignees
No one assigned
Labels
bug
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
[BUG] Jacobian with diff_method='best' and...
Notifications
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2 participants